

Project 4

Joint Limits and IK Distribution in Orthogonal 3R Robots

Format: Group of two students; simulation-based project.

Course foundations: Lectures 3, 5–7; Exercises 1 and 3.

Motivation

The mathematical workspace can overstate what a robot can execute. Joint limits remove some postures and can disconnect a region that was connected before those limits were imposed. Two target positions may each be reachable yet inaccessible from the same initial configuration without violating a limit or encountering a singularity.

Task

Overall objectives. Compare how joint limits reshape the IK distribution and feasible configuration components of selected cuspidal and noncuspidal orthogonal 3R geometries.

- **Establish two reference geometries.** Use the noncuspidal and cuspidal idealized Lecture 6 presets: $a = (a_1, 2, 1.5)$ with $a_1 = 0$ and 1 respectively, $d = (1, 1, 0)$, and $\alpha = (\pi/2, \pi/2, 0)$. Validate forward kinematics and the position Jacobian, and keep these models distinct from the exact custom URDF.
- **Build the unconstrained maps.** Plot $\det J_p$ and its zero contours in the (q_2, q_3) plane. Map the singular curves into (ρ, z) and identify the two- and four-**IK** regions. Overlay the **IK** configurations of several targets and record which sampled singularity-free component contains each solution.
- **Apply controlled limit boxes.** Compare the periodic unconstrained case with $q_2, q_3 \in [-150^\circ, 150^\circ]$ and with $q_2 \in [-90^\circ, 120^\circ]$, $q_3 \in [-120^\circ, 90^\circ]$. Keep q_1 unrestricted so that the meridian reduction remains meaningful. These are experimental limits, not manufacturer specifications.
- **Classify feasible inverse solutions.** Retain the original mathematical **IK** count and separately count limit-admissible **IKs**, which may be one or three as well as two or four. Recompute sampled configuration components after clipping by limits. Do not connect opposite plot boundaries when that would cross a physical joint limit.
- **Explain global accessibility.** For two diagnostic target pairs, compare pointwise reachability with accessibility from a selected initial component. Display component-labelled workspace coverage and validate one candidate connection where available. Refine ambiguous bottlenecks and explain the limitations of sampled connectivity claims.

Specifics and scope

Study two geometries and the three stated limit settings, giving six comparable cases. Use the same workspace domain and color conventions for all cases. Report a counterexample to equating pointwise reachability with path accessibility if one is found; a carefully explained negative search is acceptable. Collision modeling is outside the core experiment.

Use the position Jacobian J_p for the 3R task. Confirm that the meridian reduction is valid for the selected model and operational point. Count distinct real IKs modulo periodicity, then filter joint limits separately. Identify two-/four-IK regions where present and explain any absent count rather than forcing both types into every map. Show both determinant values and the zero contour, and distinguish sampled connectivity from a proof. Refine the decisive region instead of claiming completeness from a coarse grid.

Expected outcome

Distinguish mathematical IK multiplicity, limit-admissible solutions, and accessibility within one feasible component. Deliver reduced determinant plots, two-/four-IK reference maps, constrained IK-count maps, and a comparison of component-dependent workspace coverage.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The idealized D–H atlas in Lecture 6 is a separate mathematical family, implemented in `lectures_main/js/viz/cuspidalityLecture.js`. Its dimensions and conventions must not be inferred from the `custom_3R` STL renderings. Use the standard-D–H convention stated in the lectures.

The exact `custom_3R` is an orthogonal 3R robot with non-intersecting consecutive joint axes. Its URDF is `lectures_main/assets/models/custom_3R/custom_3R_new.urdf`; STL meshes are in the same directory. Use the lecture website to inspect axes and frames, verify D–H parameters, plot IKs, and animate configurations. Use the Lecture 3 IK method and Lecture 6 component visualizations; the exact custom model illustrates geometry while the stated presets define the analysis.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL’s 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.